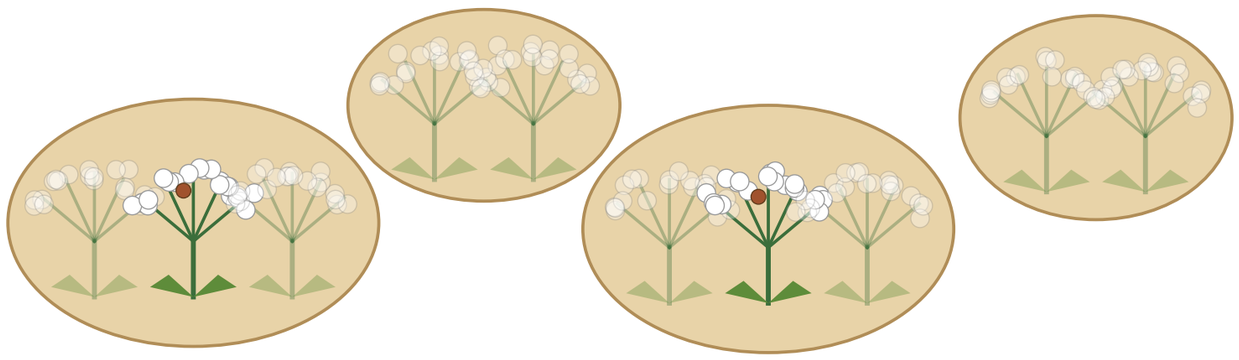
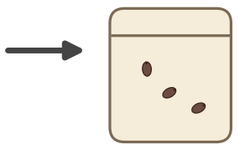
# The challenge: very little seed to start with — why?

**federally threatened (ESA)**

*wild populations: limited fertility — few fruits*





**very little seed**

✗ **far too little to restore populations directly**

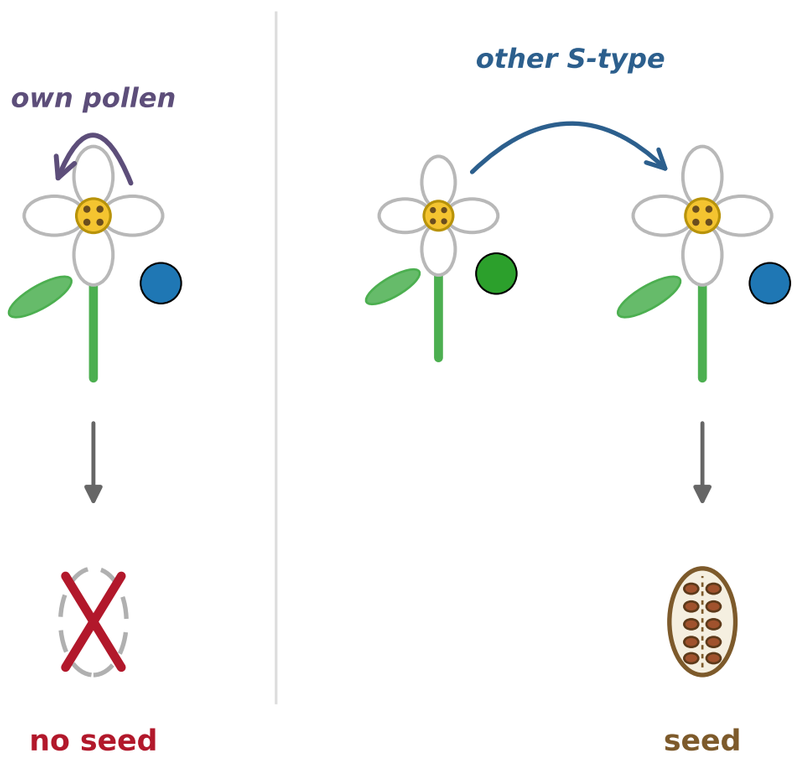
✗ **collecting more would deplete the populations it comes from**

## 1. Self-incompatible: plants reject their own pollen



color = S-type
(32 known in LEPA)

**self-incompatible**

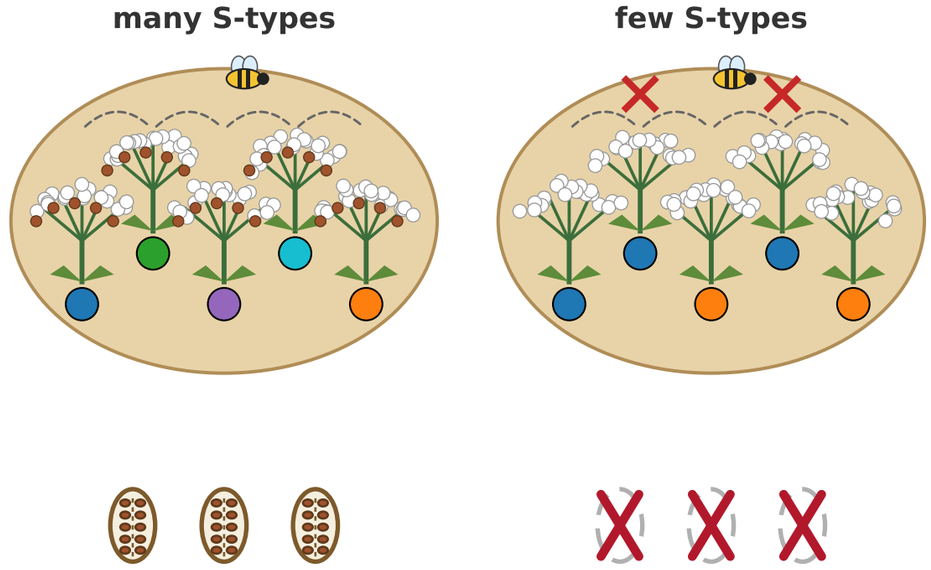



the S-locus (a male + a female gene) gives each plant an S-type: pollen with the flower's own S-type is rejected

## 2. Fragmentation + drift → mate limitation → population decline

**mate limitation**

habitat fragmentation
+ genetic drift

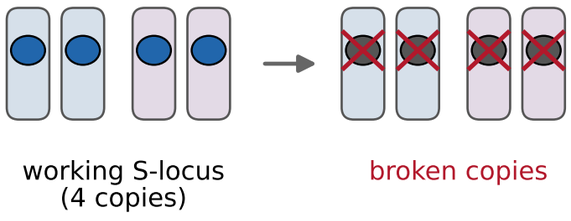



*in the field: less seed than plant size predicts*

small, isolated populations lose S-types by drift:
plants run out of compatible mates and set little seed

## 3. Rapid adaptation: escape from SI → self-compatible, short-term gain

**self-compatible**



working S-locus
(4 copies)

broken copies

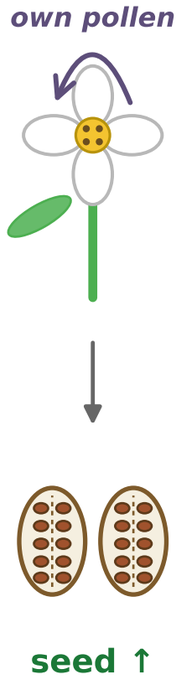
**LEPA — 263 plants genotyped**

**247** self-incompatible

**15** partly broken

**1** self-compatible

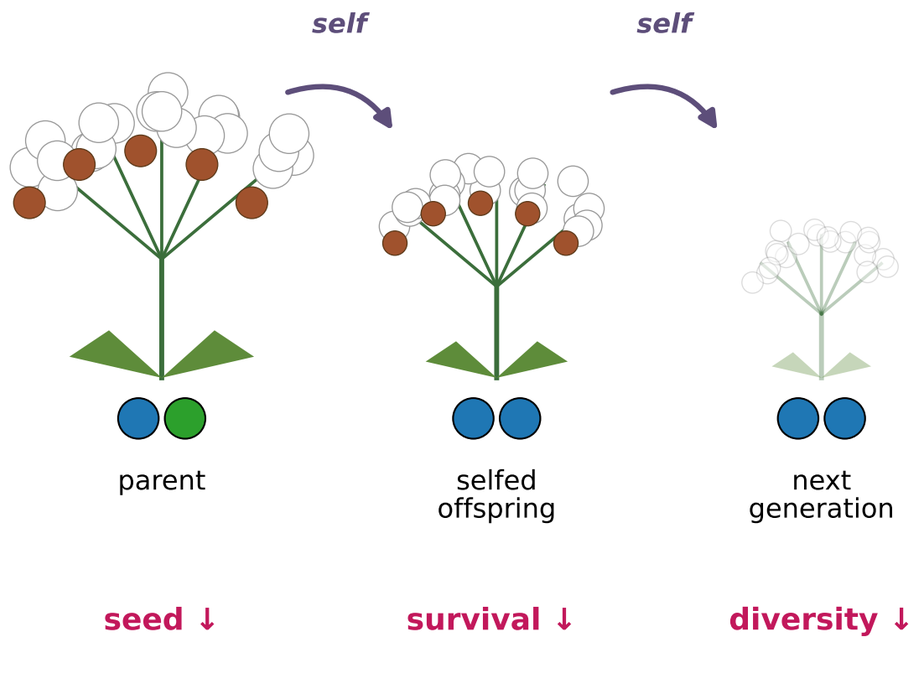
*in the field: more seed than plant size predicts*




plants with broken S-locus copies accept their own pollen and set seed alone, without mates

## 4. Long-term cost: inbreeding depression → population decline

**inbreeding**




selfed offspring grow more alike and weaker, generation after generation